

Chapter 1

INTRODUCTION

Background of the Study

Printing and copying of papers are routine functions in various industries, including education, business, and government. Their costs are usually over- or under-estimated, resulting in wasteful spending and unnecessary expenditures. Traditional cost estimating is typically based on a fixed price contract that ignores the actual composition of the document, such as the text-to-image ratio. By applying a Pixel-Based Algorithm, organizations can obtain an improved and more realistic cost estimation system to optimize their resource utilization.

Some research has shown technological advancements concentrating on cost estimation and print efficiency. (Li et al., 2024) Discuss the advantages of Aerosol Jet Printing (AJP) in achieving high-resolution printing on personal electronic devices, but note that there are disadvantages related to controlled electrical performance only. (Montgomery et al., 2023) Discuss the rapidly changing wave in additive manufacturing (AM), i.e., 3D printing, which integrates quality, material efficiency, and speed, thereby becoming a stronger contender for commercialization. (James et al., 2022) present the growing need to verify image authenticity for forensic and legal applications, thus requiring advanced image detection methods. (Lu et al., 2023) Highlight the benefits

of additive manufacturing (AM) over conventional manufacturing, where reduced costs are supported by facilitating resources and design freedom. (Krachmalnicoff & Tomasi, 2019) Depict machine learning algorithms employed in handling big data and how they demonstrate that automated systems have been able to exhibit greater efficiency and decision-making. The study favors the use, cost reduction, and application of intelligent algorithms toward improving printing and scanning.

Even though printing and scanning technology has improved, most companies are still troubled with cost estimation. The cause is that a large number of users lack a good system for calculating the true cost of scanning and printing. High-density image documents, most of the time, use more ink but pay the same as a text document. This results in overpayment and underpayment, leading to cost inefficiency. Additionally, companies are left with unnecessary paper and ink usage due to poor cost control, resulting in both unnecessary expense and environmental pollution. Without an intelligent automated system, end-users are unable to make informed decisions about document processing needs based on smart insights (Nguyen et al., 2023).

In conclusion, the use of a Pixel-Based Algorithm for Estimation will greatly improve the productivity and accuracy of scanning and print procedures. By eliminating the inefficiencies of cost estimation loss and resource wastage today, the system will ensure a reliable and effective solution for users. The objective of this research is to bridge the gap between current cost estimation technology and technology-based cost

estimations, ensuring an effective cost-saving document process system (Montgomery et al., 2023).

Statement of the Problem

While demand for effective document processing is on the rise, it is challenging to accurately estimate the cost of printing and scanning. Traditional price policies often include predetermined prices, regardless of the contents of any document, resulting in underpayment and overpayment. All resources are wasted wastefully, and unjustified money is wasted without a system of cost estimation.

The research seeks to identify and address the following problems:

- **Mis-estimation of Costs with Fixed Pricing Models** - Existing pricing models for printing and scanning do not consider the actual content in a document, such as the text-to-image ratio or color intensity. This leads to discriminatory pricing, where plain text documents are overcharged, and image documents are undercharged, resulting in cost inefficiencies.
- **Automated Failure to Distinguish Black & White and Color Documents** - Most printing systems never distinguish between black & white and color documents. This leads to mispricing, where color documents are priced as black and white, and black and white documents are charged as if they were color documents. Organizations lack cost fairness and transparency when there is

no automated classification system in place to categorize documents.

- **Inefficient and Time-Consuming Cost Estimation Process -**

The cost estimation process is cumbersome and prone to errors when executed manually. Users cannot accurately estimate the cost of scanning and printing and thus pay unwanted fees. A Pixel-Based Algorithm-integrated Optical Character Recognition technology must be applied to make the cost estimation process efficient and effective.

Objectives of the Study

The overall goal of this study is to design and develop a Printing and Scanning Cost Estimation Application. More specifically, the goal of this study is:

- **Integrate a Pixel-Based Algorithm for Cost Estimation -**

Integrate a Pixel-Based Algorithm to a document in order to estimate true printing and scanning costs.

- **Utilization of Thresholding Algorithm for the Classification of Colored & Black & White Document -** to be priced correspondingly because color printing would be pricier in most cases than black & white.

- **Scan Cost Calculation -** This technology calculates print and scan costs precisely using pixel-based formulas and optical character recognition algorithms. It transforms the image-based

and text-covered document, ensuring an equitable and fair price display.

Significance of the Study

The research discussed above is essential for transforming cost management in organizations that handle vast volumes of documents. Compared to the limitations of classical flat-rate cost estimating, pixel-based algorithms, which constitute the crux of the research, are more precise and faster methods of isolating color from non-color content. This leads to efficient cost allocation, maximum resource utilization, and overall cost reduction.

Businesses and organizations. Can gain an advantage from the research findings by reducing the cost incurred for scanning and printing, increasing overall resource management, and being cost-effective with pixel-based cost estimation.

IT and Print Management Professionals. The research provides evidence-based suggestions that can be used to enhance existing document processing systems, enabling professionals to implement more accurate resource management and billing procedures.

Researchers and Scholars. The research for this paper provides a template for designing pixel-based cost estimation algorithms, and it calls for further research in cost estimation for document processing and new areas of investigation.

Environmental Sustainability Benefactors. By minimizing unwanted printing and streamlining efficiency, the paper indirectly promotes sustainability by reducing paper and ink wastage. The research discussed above is essential for transforming cost management in organizations that handle vast volumes of documents. Compared to the limitations of classical flat-rate cost estimating, pixel-based algorithms, which constitute the crux of the study, are more precise and faster methods for isolating color from non-color content. This leads to efficient cost allocation, maximum utilization of resources, and overall cost reduction.

Scope and Delimitation of the Study

Scope

The proposed work is to maximize printing and scanning expenses based on pixel algorithms to improve the estimation of document content and estimate real-world samples at a reduced cost. The system uses significant functionalities like scanning, pixel-by-pixel analysis, real-time cost estimation, and reporting. The system is suggested to help small and medium businesses (SMEs) minimize print and scan expenses by classifying color and non-color content, monitoring resource use, and estimating the exact cost per process operation of each document process. The system can:

Document Processing Capabilities

- Upload and scan documents for processing.
- Processed by pixel for computing color use.
- Estimate real-time print costs based on perceived color intensity.
- Produce cost reports.

Admin

- Add, change, and remove accounts (system users).
- Cost configurations can be configured to print and scan.
- Print and email cost reports.
- Display system activity logs.

Users

- Scan and upload files.
- Display the approximate cost of printing.
- Print your cost reports locally.

Location

Agusan del Sur State College of Agriculture and Technology
(ASSCAT), San Teodoro, Bunawan, Agusan del Sur.

Delimitation

The study limitations are as follows:

- There is no interconnectivity with third-party accounting applications, therefore not for complete financial management purposes.

- It accepts standard document input only, such as PDF, PNG, and JPEG, but not less commonly used file extensions.
- It is limited to pixel-based color detection and lacks high-end image processing algorithms like optical character recognition (OCR) or pattern recognition.
- It is designed for small and medium-sized businesses, not large commercial print companies needing high-end industrial applications.

Time and Place of the Study

The research will be conducted in San Teodoro, Bunawan Agusan del Sur (ASSCAT). The machine will be fabricated at Agusan del Sur State College of Agriculture and Technology (ASSCAT). Machine testing will be commended in March 2025 and end in May 2025.

Conceptual Framework

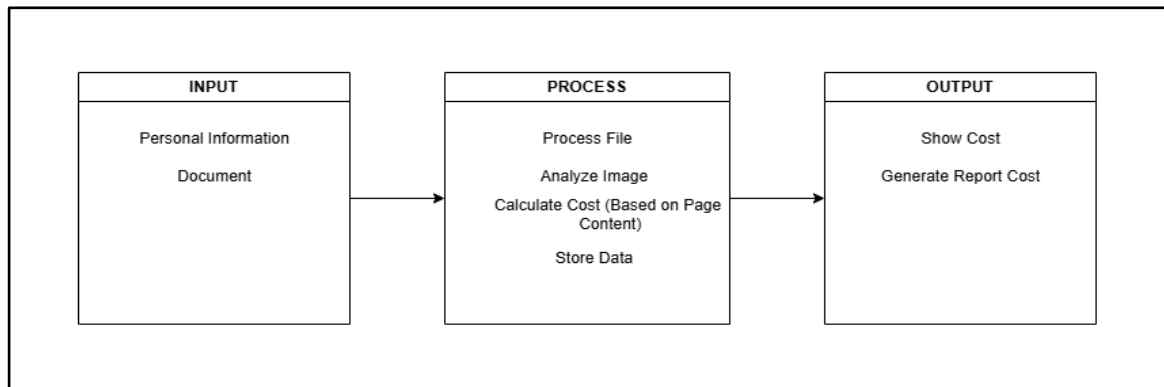


Figure 1. Conceptual Framework of the Study

The image depicts a data flow diagram with three main steps: Input, Process, and Output. In the Input process, sensitive data and files are entered into the system. The pillow is used here to read and open image files to edit and resize them quickly. Advanced methods are used after the data has been uploaded to the Process process. OpenCV works on heavy image processing, such as processing the document content, and RGB Thresholding is performed first to separate proper color channels such as Red, Green, or Blue. Adaptive Thresholding and Otsu's Thresholding are used to effectively mark text or objects in the image by thresholding it into gray and finding the optimal segmentation. Gray scaling is generally done to make this simple. While this is being computed, NumPy assists in the calculation in mathematics in an attempt to determine the costs depending on the content of the pages. All the information needed is saved for future use. The system returns the calculated cost and a cost report at the Output stage. Locally on the server, XAMPP serves the system for database interaction reasons. Django Filters ensures the sanitizing and reformatting of the results by removing any excessive or out-of-range values before

display. All this guarantees cost estimation and reporting accuracy according to document content.

Operational Definition of Terms

Pixel-Based Algorithm. A Software used in searching pixels one by one in an image to perform operations such as cost calculation, color identification, and image separation to reach maximum scanning and printing.

Thresholding. An operation of converting grayscale images to black and white images by assigning a specific value to distinguish background and foreground, whose use is recognition of the area of printing.

Grayscale Conversion. The conversion of color images into shades of gray simplifies the analysis by removing image complexity and prominences.

Pillow Library. This low-level Python library processes image-processing operations such as opening, reading, editing, resizing, and saving images.

OpenCV Library. This low-level image processing library can handle high-level operations such as filtering, edge detection, Thresholding, and real-time image analysis.

RGB Thresholding. is a color filter that distinguishes separate color channels (Red, Green, Blue) from an image to look for color usage, which can influence printing costs.

Adaptive Thresholding. A Threshold operation with different threshold values on different regions of an image is applied whenever the light condition is inhomogeneous in the image.

Otsu's Thresholding. is an Automatic thresholding algorithm that calculates the best threshold value to differentiate between foreground and background without human intervention.

XAMMP. Local server installation through which programmers can run backend programs and databases for testing and development.

Image Processing. A process of performing things on images to obtain data, enhance images, or transform them for a specific purpose, such as printing or scanning.